

Problem Set 1

Dmitry Mukhin

d.mukhin@lse.ac.uk

Problem 1

Suppose that an agent gets utility not only from consumption over T periods of her life, but also from leaving a bequest to its descendants

$$\sum_{t=0}^T \beta^t \log(C_t) + \eta \beta^T \log B_{T+1}$$

and is subject to a standard budget constraint

$$B_{t+1} = RB_t + Y_t - C_t.$$

1. Is the additional constraint $B_{T+1} \geq 0$ discussed in the lecture binding in this case?
2. Derive the optimality conditions and provide intuition for the terminal condition.
3. Solve for the optimal consumption C_0 and bequest B_{T+1} assuming $B_0 = 0$ and $Y_t = Y$.
4. What happens to the optimal bequest when $T \rightarrow \infty$ if $\beta R = 1$? Explain.

Problem 2

Consider an agent with a CRRA utility

$$\int_0^T e^{-\rho t} \frac{C_t^{1-\sigma} - 1}{1-\sigma} dt$$

and a budget constraint

$$\dot{B}_t = rB_t - C_t,$$

where labor income is zero, $r = \rho$, and $B_0 > 0$.

1. Show that the CRRA utility converges to $u(C) = \log C$ in the limit $\sigma \rightarrow 1$. Focus on this limit for the rest of the problem.
2. Write down the Hamiltonian and derive the optimality conditions.
3. Obtain the intertemporal budget constraint and solve for the optimal consumption.
4. Suppose the agent can choose not only consumption path, but also the length of life T . Assuming utility is zero for $t > T$ and the agent cannot leave any debt, solve for the optimal value of T .
5. Derive the optimality condition for T using a perturbation argument.

Problem 3

Consider the following investment problem. Given initial value of capital K_0 , a firm chooses investment path $\{I_t\}_{t=0}^{\infty}$ to maximize profits

$$\sum_{t=0}^{\infty} \beta^t \left[A_t K_t^\alpha - I_t - \frac{\phi}{2} \frac{I_t^2}{K_t} \right]$$

subject to the capital law of motion

$$K_{t+1} = (1 - \delta)K_t + I_t,$$

where A_t is the firm's productivity, $\alpha \in (0, 1)$, $\phi > 0$ stays for the capital adjustment costs, $\delta \in (0, 1)$ is the depreciation rate.

1. Write down the Lagrangian and take the first-order conditions.
2. Use a finite-period version of the model to derive the transversality condition.
3. Derive the analog of the Euler equation, i.e. the optimality condition for capital and investment that does not include λ_t . Interpret this equation using the perturbations of the optimal path.
4. Rewrite the capital law of motion for an arbitrary length of period Δ . Obtain the continuous-time version of the optimization problem.

5. Define the Hamiltonian and write down the optimality conditions.
6. Show that the same continuous-time optimality conditions can be obtained directly from the optimality conditions in discrete time.